Early Detection of Wheat Diseases

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ABSTRACT:

Wheat is one of the important crop in the field of agriculture, but it can easily vulnerable to diseases that effect its yield and quality. This Early detection of wheat diseases plays crucial role in improving productivity and reducing the loss. This research paper uses deep learning-based approach for identifying wheat diseases using image classification techniques.

The proposed system uses transfer learning model based on EfficentNet-B0 to classify leaf images of wheat into multiple categories. The used dataset consists different diseases classes like Aphid, Rusts, Mildew, Septoria, Healthy leaves, including other nine classes. The dataset is preprocessed using resizing, normalization and augmentation techniques

The model trained and evaluated using performance metrics such as accuracy, confusion matrix and per class accuracy. This proposed system achieved good accuracy , with better performance for some classes compared to others The system demonstrates the feasibility of applying deep learning techniques for agricultural disease detection and can be further improved for the real-world deployment.

Keywords:

Deep learning, Transfer Learning, Wheat Disease Detection, Image classification, Agriculture

I. INTRODUCTION

Agriculture plays vital role in sustaining human life and wheat is one the mostly grown crops in worldwide. However, wheat plants have natural tendency to catch several diseases like Brown rust, Yellow rust, Mildew and Septoria which can significantly reduce crop yield and quality.

Some of the other common wheat diseases that we considered are, Aphid, Black rust, Blast, Fusarium head blight, Leaf blight, Mite, Smut, Stem fly, Tan spot.

Traditional disease detection methods mainly focused on manual observations, which consume more time and they require expert knowledge. So, with the advancement in artificial intelligence, deep learning has emerged as most powerful tool for image-based classification tasks.

This project mainly focuses on developing deep learning-based system which automatically detect the wheat disease from leaf images. This system helps farmers in identifying diseases at early stages and can take step to prevent it.

II. LITERATURE REVIEW

By analyzing the previous studies based on deep-learning we get to know that deep-learning techniques can effectively identify the plant diseases. CNN based models are widely used for feature extraction and classify the plant images. Researchers have also used transfer learning models like MobileNet, ResNet and VGG to improve performance.

Recently, EfficientNet models has been introduced, which provide better accuracy and efficiency compared to other models. These models are trained on large datasets and that can be easily adaptable for specific tasks like plant diseases detection.

This study builds upon existing work by implementing a CNN based classification system and evaluating its performance on wheat disease detection.

III. METHODOLOGY

A. System Overview

The system follows a structured pipeline:

- Image input

- Image Preprocessing
- Model training
- Prediction output

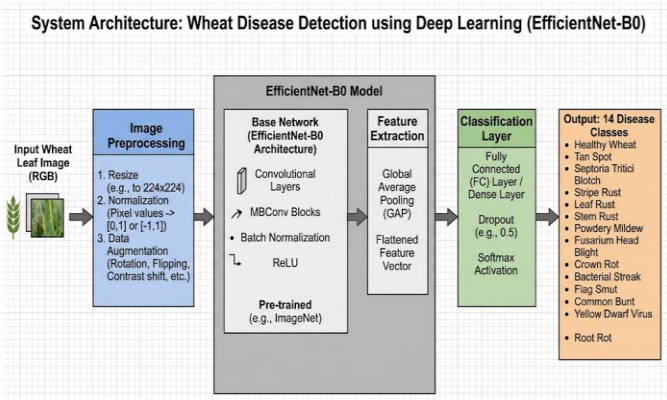


Fig. 1: System Architecture

B. Dataset Description

The dataset consists of wheat images categorized into 14 classes:

- Brown Rust
- Healthy
- Mildew
- Septoria
- Aphid
- Smut
- Black Rust etc., many others…

The dataset is split for training and testing purposes.

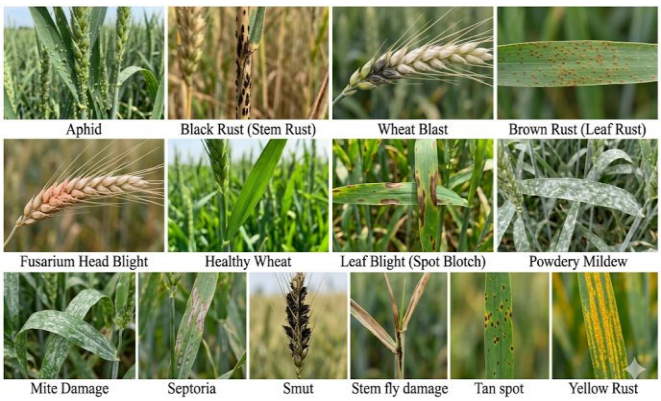


Fig 2: Sample Dataset Images

C. Data Preprocessing

The following preprocessing steps were used

Resizing images to 224*224

Normalization

Data augmentation (rotation, flipping)

These steps help to improve model performance and reduce overfitting.

D. Model Development

In this project, a transfer learning approach is used with EfficientNet-B0

- The model is pretrained on ImageNet
- Feature extraction layers are initially frozen
- Final classification layer is modified for 14 classes
- Fine tuning is applied after initial training

The model helps to extract features from input images and classify them into their correct disease categories.

E. Training Details

- Optimizer: Adam
- Loss function: Categorical Crossentropy
- Epochs:100
- Dataset Size:32 images

Class weights are used to handle imbalance in dataset

F. Evaluation Metrics

The model performance is evaluated using

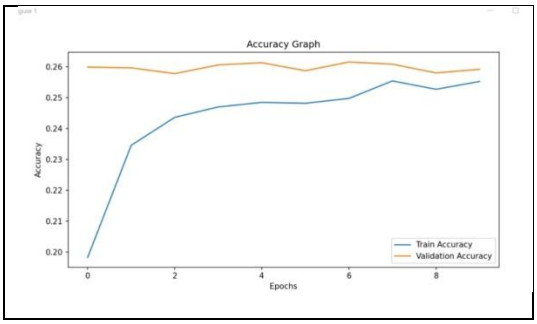
Accuracy

Confusion Matrix

Per Class Accuracy

IV. RESULTS AND DISCUSSION

A. Accuracy and Loss Graph



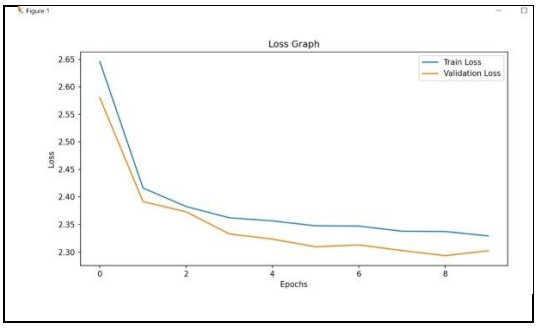


Fig 3: Training and validation accuracy and loss over epochs

Observation: The training and validation accuracy increased over epochs, while loss decreased steadily. This shows model is learned properly.

B. Confusion matrix

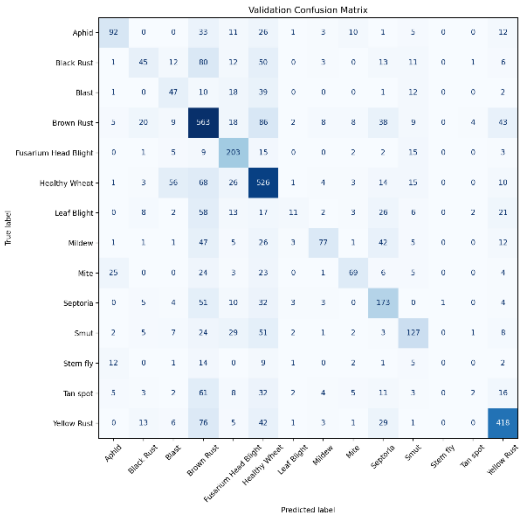


Fig 4: Confusion matrix showing classification performance

Observation:

Confusion matrix shows class wise performance. Almost many classes predict correctly, but some confusion arises between similar classes, ex, rust types.

C. Discussion

The result indicate that the model performs well for certain disease categories, but not equally for all. Diseases which look similar are tougher to classify. Increasing dataset size and improving image quality can help to improve results.

Limitations:

There are some limitations that can able to influence the overall performance of the model.

Firstly, the dataset which is used for training is only limited for some classes, which is restricts the model's ability to learn diverse patterns.

Secondly, there is a imbalance in the number of images across the different classes, which can lead the model to favor certain categories over others.

Additionally, some diseases share very similar visual characteristics, which cause the reason for model to clearly distinguish between them and Training done on CPU is slower.

Even with these limitations, the model shows that deep learning can be useful in agriculture applications, especially in the context of plant disease identification.

V. CONCLUSION AND FUTURE WORK

This study presented the system based on Deep learning based system for detecting disease of the wheat using image classification methods. The EfficientNet-B0 model shows better performance compared to basic CNN model. The model achieved the accuracy around 67-70%, which shows that it is capable of identifying plant disease to some moderate extent.

Such a system could be quite helpful for farmers, as it enables early detection of crop diseases, which can help to reduce losses and improve productivity. However, the current model still has many limitations, and need further improvements to enhance the reliability and performance under different real-world conditions.

Future enhancements include:

Increasing the size and diversity of the dataset which can allow the model to learn better and generalize more effectively. The use of advanced transfer learning models such as Vision transformers could also helps to improve the accuracy and efficiency.

In addition, developing a user-friendly mobile application would make the system more accessible to farmers. Another important invention can be done is enabling real time detection using a camera and deploy model on cloud and also we can try to extend to other crops such as rice, ragi etc as well, which can make solution easier and more practical to use in field conditions.

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